

# MiniMines Sales Hub — Product Description Document

**Version:** Engine v2 (2026-07-16)

**Repository:** [codeunderscrap/attplatform](https://github.com/codeunderscrap/attplatform)

**Stack:** Python (FastAPI) · React/TypeScript (Vite) · PostgreSQL / SQLite · Docker

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## 1. What Is MiniMines Sales Hub?

MiniMines Sales Hub is an **AI-powered internal sales intelligence platform** for a chemical and battery materials trading company. It ingests raw EXIM (Export-Import) trade data files, runs a multi-stage scoring pipeline, and surfaces ranked shortlists of:

- **Chemical trading opportunities** — which chemicals in MiniMines' portfolio have the highest Attractiveness-to-Trade (ATT) score, and which new chemicals detected in import data are potential expansion opportunities.
- **Battery material procurement intelligence** — which global suppliers and buyers of battery scrap/materials are Tier A/B/C procurement targets.
- **EPR (Extended Producer Responsibility) cross-links** — which companies are importing materials relevant to their EPR obligations and are therefore high-priority sales contacts.

The platform is used daily by the MiniMines sales team to prioritize outreach, validate trading theses, and track market trends — replacing manual spreadsheet analysis.

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## 2. Core Modules

### 2.1 Chemical ATT Pipeline (*ATT = Attractiveness to Trade*)

Analyzes every chemical against 8 dimensions of trading attractiveness and produces a score 0–100.

#### Inputs:

- One or more EXIM trade data `.xlsx` files (shipment rows: date, HSN code, seller, buyer, qty, unit, value, description)
- A "base portfolio" `.xlsx` file listing the chemicals MiniMines already trades

#### Outputs:

- ATT score per chemical (0–100, Tier A/B/C)
- Ranked table: base-portfolio chemicals + opportunity chemicals
- 8-dimension breakdown per chemical

- Time trend chart (growing / stable / declining)
- Buyer & supplier intel tables
- Geopolitical risk log
- Regulatory risk flag
- India incentive bonuses (RoDTEP, drawback)
- Downloadable Excel report with methodology sheets

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## **2.2 Battery Procurement Module**

Analyzes battery material (scrap, cells, modules) import trade data to rank global suppliers and buyers.

### **Inputs:**

- EXIM `.xlsx` file(s) for battery-related shipments

### **Outputs:**

- Entity list (suppliers / buyers) ranked by procurement attractiveness (Tier A/B/C)
- Per-category (Li-ion, Lead-acid, E-waste, etc.) price benchmarks
- Consistency, volume, geo-ease, and price-index scores per entity
- Downloadable Excel report

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## **2.3 EPR Intelligence Module**

Cross-links the ATT chemical database with companies' EPR (Extended Producer Responsibility) target obligations.

### **Inputs:**

- Configured EPR targets (material → tonnes obligation) per company
- ATT run database (shipment history)

### **Outputs:**

- Company priority score based on:  $\text{EPR gap (target - credits)} \times \text{material importance weight}$
- Cross-linked shipment history: which companies have imported ATT chemicals from recent EXIM runs
- AI-generated company research briefs and deal memo drafts (via LLM integration)

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## **2.4 Leads & CRM Module**

Lightweight CRM layer for the sales team.

## Features:

- Lead pipeline with stages: Prospect → Qualified → Proposal → Won/Lost
- Notes and activity log per lead
- Link leads to ATT chemicals or EPR-identified companies
- API-key-gated access for external integrations

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## 2.5 Settings & Administration

- Configurable dimension weights (volume, price, buyers, suppliers, trend, structure, freedom, barrier)
- Configurable tier thresholds (Tier A  $\geq 70$ , Tier B  $\geq 40$ )
- Anchor-band floor/ceiling for log-scale dimensions (admin-editable per dimension)
- EPR material weights
- LLM provider configuration (API key, model)
- PIN-gated access (rate-limited, hashed at rest)
- Retention policy (default: keep forever; opt-in auto-delete with confirmation)
- Trend exclusion list (months to exclude from regression, e.g. partial current month)

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## 3. User Flows

### *Upload & Score a Chemical Run*

1. Go to **Upload** → select EXIM file(s) + (optional) custom base portfolio
2. Platform warns if the exact file was previously uploaded (duplicate detection)
3. Pipeline runs in the background with a live progress bar
4. On completion, navigate to **Dashboard** → select the run → view ranked chemicals

### *Explore Battery Run*

1. Go to **Battery** tab → upload EXIM battery files → pipeline runs
2. View supplier/buyer entity tables, filterable by role and tier
3. Download Excel report for offline sharing

### *EPR Prioritization*

1. Go to **EPR** tab → see companies ranked by Priority Score

2. Click **AI Research & Deals** to generate a company brief via LLM
3. Cross-links show which recent EXIM runs contain that company's shipments

### ***Trader Feedback Loop***

1. On the **Rankings** page, traders can vote on a chemical: **Confirm / Challenge / Correct**
2. **Challenge** → −2 pts ATT adjustment on next score
3. **Correct** → ±2.5 pts depending on whether the suggested tier is above/below the current score
4. **Confirm** → 0 pts (displayed but not scored — prevents gaming the system)
5. Feedback persists across run deletions — scores stay calibrated over time

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## **4. Technology Stack**

| Layer            | Technology                                                    |
|------------------|---------------------------------------------------------------|
| Backend API      | Python 3.11, FastAPI, Uvicorn                                 |
| Database         | PostgreSQL (production) / SQLite WAL (development)            |
| ORM              | SQLAlchemy                                                    |
| Data Processing  | openpyxl, pandas-free (custom row parser)                     |
| NLP Matching     | Custom token-similarity (SequenceMatcher) + optional LLM      |
| LLM Integration  | OpenAI / configurable provider                                |
| Frontend         | React 18, TypeScript, Vite                                    |
| UI State         | React hooks (no external state library)                       |
| Containerization | Docker + docker-compose                                       |
| Auth             | PIN (rate-limited, bcrypt-style SHA256+salt) + API-key header |

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## **5. Deployment**

ATT\_Platform/

- backend/ # FastAPI app, pipeline, DB models
- ■■■ main.py # API routes, auth, upload handling
- ■■■ db.py # SQLAlchemy models, migrations
- ■■■ runner.py # Background job orchestration
- ■■■ epr.py # EPR cross-link logic
- ■■■ leads.py # CRM routes

- ■■■■ settings.py # Settings CRUD, PIN hashing
- ■■■■ rescore.py # Batch re-score utility (engine upgrades)
- ■■■■ pipeline/
- ■■■■ engine.py # Chemical ATT stages 1-6
- ■■■■ battery.py # Battery scoring pipeline
- ■■■■ constants.py # Weights, anchor bands, geo events, regulatory DB
- ■■■■ battery\_export.py # Excel report generator
- frontend/ # React/Vite SPA
- ■■■■ src/pages/ # Dashboard, Upload, Battery, EPR, Rankings, Settings...
- data/ # SQLite file + uploaded EXIM files
- Dockerfile
- start.bat # Windows one-click launcher
- requirements.txt

Quick Start (Windows)

```
double-click start.bat

# or

venv\Scripts\activate

uvicorn backend.main:app --host 0.0.0.0 --port 8000
```

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6. Key Design Decisions

|         | Rationale                                                                                                                                           |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| v2)     | Cohort-independent — adding/removing one chemical doesn't shift any other chemical's score. Identical raw values score identically.                 |
|         | Same shipment appearing in two overlapping EXIM uploads counts once in cross-run dashboards, not twice.                                             |
| on      | Trader domain knowledge accumulates over time and should not silently disappear when old runs age out.                                              |
| revert) | Auto-deleting a sales team's run history by default is the wrong default. Deletion requires explicit opt-in with a confirmation dialog.             |
|         | Prevents a coordinated "confirm" campaign from artificially inflating scores beyond what the data supports.                                         |
|         | A 4–6 digit PIN's tiny keyspace means hashing alone can't stop brute force — rate limiting is the real defense; hashing only protects against a cop |

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7. Upcoming / Planned

- EPR engine redesign (Q1–Q10 in `docs/EPR\_ENGINE\_PROBLEMS.md`) — awaiting design sign-off

- NLP sub-categorization of "Battery Scrap (Mixed/Other)" descriptions (R4b extension)
- Full cross-run movers dashboard (biggest ATT score changes between runs)
- Mobile-responsive UI

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\*Document generated 2026-07-16 from Engine v2 implementation. For scoring math, see the companion Backend Logic document.\*